Department of Physics
PCS 125 – Physics: Waves and Fields

Please insert in the circle the initial of your family name

Please Print Clearly:

Student:_____ _____
Last Name *First Name*

Student Number: _____ **Student Section:** _____

Version B – Please bubble 'B' under version on the bubble sheet

Midterm Test – March 10, 2022

Read the following attentively

- **Ensure that your name and student number are correct in the bubble sheet.**
- **Duration of the exam is 80 minutes.**
- **The exam consists of 12 multiple-choice questions. Answer ALL questions and show all work.**
- **Each multiple-choice question is worth 1 mark.**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer. If no work or correct argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.**
  - **The work shown in your exam booklet may be checked, at least for one of the questions, selected randomly.**
  - **Transfer your choices to the bubble sheet only when you are sure of them.**
  - **Some answers may be rounded off.**
- **Pagers, cell phones, watches, and other electronic devices (other than a calculators fitting the description below) must be silenced and placed out of reach. Hats and earphones are not permitted. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.**
- **Sharing of pencils, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **The calculator models allowed for the test are either Sharp EL-546 or Casio FX-991 (or similar models). If you are found to have another type of calculator, it may be confiscated for the duration of the test.**
- **The exam booklet has a total of 8 pages. You are responsible for checking that you have all pages.**
- **You may not leave the exam room during the final 15 minutes of the exam.** You may detach the formula sheet on the last page.

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Sign here to signify you read the instructions

1. As a certain sound wave travels through a kind of gas (density 1.20 kg/m^3), it produces pressure variations (above and below certain pressure) given by $\Delta P = 2.54 \sin [\pi x/2 + 200\pi t + 3\pi/4)]$, with all quantities in SI units. What is the displacement amplitude of the wave?

(a) 4.21×10^{-6} m

(b) 1.68×10^{-5} m

(c) 1.01×10^{-5} m

(d) 8.42×10^{-6} m

(e) 9.82×10^{-6} m

2. For an object undergoing simple harmonic motion,

(a) the magnitude of the acceleration is greatest when the speed is greatest.

(b) the total mechanical energy is proportional to the amplitude.

(c) the magnitude of the displacement is greatest when the speed is greatest.

(d) the maximum potential energy is larger than the maximum kinetic energy.

(e) the magnitude of the acceleration is greatest when the magnitude of the displacement is greatest.

3. The wave function for a traveling wave on a taut string is (in SI units)
$y(x,t) = 0.370 \sin(2\pi x - 8\pi t + \pi/4)$. What is the maximum transverse speed of an element of the string?

(a) 25.1 m/s

(b) 0.25 m/s

(c) 9.30 m/s

(d) 2.32 m/s

(e) 4.00 m/s

4. There are two speakers separated by a distance of d. The speakers both emit sound waves, in phase, with a wavelength λ. Point P is a distance L_1 from speaker 1 and a distance L_2 from speaker 2. What can you say about the distances L_1 and L_2 if the sound emitted by the two speakers experiences destructive interference at at point P?

(a) Distance d is equal to $|L_2 - L_1|$

(b) The difference in distances from the speakers to point P is equal to an integer number of the speaker separation distance: $|L_2 - L_1| = nd$, where n is a positive integer.

(c) The difference in distances from the speakers to point P is equal to an integer number of wavelengths: $|L_2 - L_1| = n\lambda$, where n is a positive integer.

(d) The difference in distances from the speakers to point P is equal to an integer + one half number of the speaker separation distance: $|L_2 - L_1| = (n + 1/2)d$, where n is an integer greater than or equal to zero.

(e) The difference in distances from the speakers to point P is equal to an integer + one half number of wavelengths: $|L_2 - L_1| = (n + 1/2)\lambda$, where n is an integer greater than or equal to zero.

5. A violin is played, producing sound with an initial intensity I_i that increases to a final intensity I_f. If the final intensity is 10 times the initial intensity ($I_f = 10\ I_i$), what is the difference in sound level between the final and initial sounds produced by the violin?

a) 1dB

b) 100dB

c) 2dB

d) 10dB

e) None of the above

6. There are two waves, described by
$y_1(x, t) = 8.00\text{ cm} \sin(kx - \omega t)$, and
$y_2(x, t) = 8.00\text{ cm} \sin(kx + \omega t)$.
The propagation velocity of these waves is v = 85.7 m/s and the period is T = 0.210 s. The two waves are summed together, $y_{sum}(x, t) = y_1(x, t) + y_2(x, t)$.
For $y_{sum}(x, t)$, what is the amplitude of the simple harmonic oscillations at x = 7.00 m?

(a) 10.3 cm

(b) 0.682 cm

(c) 11.3 cm

(d) 13.9 cm

(e) 16.0 cm

7. Which of the following frequencies could *NOT* be present as a standing wave in an organ pipe open at both ends? The fundamental frequency is 85 Hz.

(a) 85 Hz

(b) 382 Hz

(c) 340 Hz

(d) 340 Hz

(e) 170 Hz

8. A long string is constructed by joining the ends of two shorter strings. The tension in the strings is the same but string I has 4 times the linear mass density of string II. When a sinusoidal wave passes from string I to string II:

(a) the frequency increases by a factor of 2

(b) the frequency decreases by a factor of 2

(c) the wavelength decreases by a factor of 2

(d) the wavelength increases by a factor of 2

(e) the wavelength stays the same

9. A 1.0 kg block oscillates on a spring with the spring constant 20 N/m. Right is the positive direction. At t = 0s the block is 20 cm to the right of the equilibrium position and moving to the left at a speed of 100cm/s. Determine the phase constant of the oscillation.

(a) 2.056 rad

(b) 0.841rad

(c) 0.020 rad

(d) 0.999 rad

(e) 0.437 rad

10. Which of the following statements is true for a standing wave in an organ pipe that is open at one end and closed at the other end?

(a) The closed end of the organ pipe is a pressure antinode

(b) The open end of the organ pipe is a displacement node

(c) The open end of the organ pipe is a pressure antinode

(d) The closed end of the organ pipe is a pressure node

(e) The closed end of the organ pipe is a displacement antinode

11. A 5.0-kg block hangs from a spring with spring constant 2000N/m. The block is pulled down 5.0 cm from the equilibrium position and given an initial velocity of 1.0 m/s back toward equilibrium. What is the total mechanical energy of the motion?

(a) 0.5 J

(b) 2.5 J

(c) 3.5 J

(d) 7.1 J

(e) 5.0 J

12. A source emits sound with a frequency of 440 Hz and is moving in a straight line with a speed of 24 m/s. An observer driving a car at a speed of 24 m/s is chasing the source along the same line. Assume the speed of sound is 343 m/s. What frequency will be heard by the observer?

a) 350Hz

b) 440 Hz

c) 230Hz

d) 610Hz

e) 550Hz